Mahatma Gandhi University Kottayam

MGU-UGP (HONORS) Regular Examination October 2025

Third Semester

Discipline Specific Elective (DSE) MG3DSEEC0200

Introduction to Data Analysis for Economics

(2024 Admission onwards)

Scheme

Duration 1 Hour 30 minutes Maximum Marks 50

Part A

Answer all questions. Each question carries 1 mark

1.b

2.b

3.b

4.b

5.b

6.a

7.b

8.c

9.a

10.d

11.b

12.c

13.b

14.c

15.b

(1X15=15 marks)

Part B

Short Answer

Answer any 10. Each question carries 2 marks

16. The CHAR() function returns a character based on the ASCII code.

Example: CHAR(65) → "A"

17. (**Liberal Valuation)**.

Can use the IF logical function in Excel to calculate the bonus based on the sales value.

Formula: =IF(Sales > 10000, Sales * 10%, 0)

Explanation:

- Condition: Sales > 10000
- If True: Bonus = 10% of Sales (Sales * 10%)
- If False: Bonus = 0

Example:
If cell A1 contains the sales value, the formula would be:

=IF(A1 > 10000, A1 * 10%, 0)

18. Discrete data are countable numerical values that occur in finite or distinct steps, not in a continuous range.

19. Simple Interest and Compound Interest are two methods of calculating the return on a loan or investment. Simple Interest is calculated only on the original principal amount throughout the entire period. For example, if ₹10,000 is invested at 10% annual interest for two years, the simple interest earned will be ₹2,000, and the total amount becomes ₹12,000. In contrast, Compound Interest is calculated on the principal as well as on the accumulated interest from previous periods. Using the same example, the compound interest after two years would be ₹2,100, making the total amount ₹12,100. Thus, compound interest grows faster than simple interest because it allows interest to earn further interest over time

20. Add-ins in data analysis tools (like Microsoft Excel) are optional programs or features that extend the functionality of the main software. They provide specialized tools for advanced analysis, automation, or visualization that are not included by default.eg.Analyis ToolPak, Solver Add-in.

21. Steps to Create a Named Cell:

- Select the cell (for example, B2) that you want to name.
- Click in the Name Box — located to the left of the formula bar.
- Type your desired name (e.g., Bonus Rate).
- Press Enter.

Now, instead of referring to the cell as B2, you can use Bonus Rate in your formulas.

Example:

If cell B2 contains the interest rate, name it Rate.
Then, in another cell, you can type:

22. A Column Chart and a Bar Chart are both used to compare data, but they differ mainly in orientation and purpose. A Column Chart displays data using vertical bars, with categories placed along the horizontal axis and values along the vertical axis. It is especially useful for showing changes or trends over time, such as monthly sales or yearly growth. In contrast, a Bar Chart uses horizontal bars, with categories on the vertical axis and values on the horizontal axis. Bar charts are more effective when comparing different categories or when category names are long, such as comparing sales performance

across products or regions. In short, column charts highlight trends, while bar charts emphasize category comparisons.

23. Steps in Excel: to find coefficient of variation

- Enter your data in a column (for example, cells A1:A10).
- Calculate the Mean:
  =AVERAGE(A1:A10)
- Calculate the Standard Deviation (Sample):
  =STDEV.S(A1:A10)

(Use STDEV.P if the data represents a population.)

- Calculate the Coefficient of Variation:
  =(STDEV.S(A1:A10)/AVERAGE(A1:A10))*100
- Press Enter — Excel will display the CV as a percentage.

24. A non-linear regression model is a type of regression analysis in which the relationship between the independent variable(s) and the dependent variable is not a straight line. Unlike linear regression, which involves a simple linear equation, non-linear regression involves curved relationships that cannot be represented by a simple linear equation. A non-linear regression model is one where the change in the dependent variable does not occur at a constant rate with changes in the independent variable. The parameters appear in a non-linear form.

25. (**Liberal Valuation)**
Two advantages of using spreadsheets in data analysis:

- Ease of Data Organization and Calculation
- Data Visualization and Interpretation

26. The COUNT() function counts only cells containing numbers, ignoring text or blank cells.

- Syntax:

=COUNT(range)

- Example:

If cells A1 to A5 contain the values:
10, 25, "Apple", 40, ""
Then the formula:
=COUNT(A1:A5)
will return 3, because there are three numeric entries (10, 25, and 40).
Thus The COUNT() function is used to count numeric entries.
Example: =COUNT(A1:A5) → returns 3.

27.A cost function expresses the total cost (TC) of production as a function of the output (Q):
$TC=f(Q)$
Components of Total Cost:
TC=FC+VC

Where FC = Fixed Cost (costs that do not change with output)VC = Variable Cost (costs that change with output)

28. Steps:

- Enter your data in a column or row.
- Example: Type your data in cells A1 to A10.
- Select a blank cell where you want the mean to appear.
- Type the formula:
- =AVERAGE(A1:A10)

- Press Enter.
- Excel will display the mean (average) of the values in that range.

29. (**Liberal Valuation)**

1. IRR assumes that all interim cash flows are reinvested at the same IRR, which is often unrealistic.

2. Projects with multiple changes in cash flow signs may produce more than one IRR, making interpretation confusing.

3. It may favor projects with higher percentage returns but lower total value, leading to poor decisions when comparing projects of different scales.

(2X10=20 marks)

Part C
Short Essay
Answer any 3 questions
Each question carries 5 marks

30. A scattergram (or scatter plot) is a graphical tool used to display the relationship between two quantitative variables. Each point on the chart represents a pair of values — one from each variable — allowing patterns, trends, and relationships to be visually identified.

Concept: In a scattergram, the independent variable (X) is plotted on the horizontal axis, and the dependent variable (Y) is plotted on the vertical axis. The pattern formed by the points indicates the type of relationship:

- Upward sloping: Positive correlation (as X increases, Y increases).
- Downward sloping: Negative correlation (as X increases, Y decreases).
- Scattered randomly: No correlation.

Importance:

1. Visual Understanding: It provides a clear, visual representation of how two variables are related.
2. Detects Relationship Type: Helps identify whether the relationship is positive, negative, or none.
3. Outlier Detection: Makes it easy to spot unusual data points.
4. Foundation for Further Analysis: Serves as the first step before performing regression or correlation analysis.

While a scattergram provides a visual indication of the relationship, the CORREL function gives a precise numerical value to describe its strength and direction. Together, they offer both qualitative (visual) and quantitative (statistical) insights into the relationship between variables.

31. The FIND() and SEARCH() functions in Excel are both used to locate the position of a specific character or substring within a text, but they differ in how they handle case and wildcards. The FIND() function is case-sensitive, meaning it distinguishes between uppercase and lowercase letters, and it does not support wildcard characters. For example, =FIND("A","Apple") returns 1, while =FIND("a","Apple") gives an error because the lowercase “a” is treated differently. In contrast, the SEARCH() function is not case-sensitive and supports wildcards, making it more flexible. For instance, =SEARCH("a","Apple") also returns 1, and =SEARCH("p*","Apple") works because of wildcard support. In short, FIND() is stricter, while SEARCH() is more forgiving and versatile for text searches.

32. (**Liberal Valuation**)

=A5 + D$2 means “add the value from the current row (A5) to a fixed value in row 2 of column D.”It’s useful in modeling when one term (like a rate, constant, or parameter) must stay the same across multiple calculations.

A5 → Relative reference

Both the column (A) and row (5) will change when the formula is copied to another cell.

Example: If copied one cell down, it becomes A6; if copied one cell right, it becomes B5.

D$2 → Mixed reference

The row (2) is fixed because of the $ sign before it.

The column (D) is relative and will change if copied horizontally.

Example: If copied to the right, it becomes E$2; if copied down, it remains D$2.

Meaning in Modelling:

The formula adds the value in cell A5 to the value in row 2 of column D (fixed row).
So, when you copy this formula down a column (e.g., from row 5 to 6, 7, 8...), the reference to A5 changes to A6, A7, A8, etc., but the reference to D$2 remains the same (always row 2).

33. 1. ₹621.43.
    2. ₹550
    3. New total = 4350−1200=₹3,1504350 - 1200 = ₹3,1504350−1200=₹3,150.

New mean =3150÷6=₹525.00.= 3150 ÷ 6 = ₹525.00.=3150÷6=₹525.00.

New median (6 observations) = average of 3rd and 4th values: (500 + 550)/2 = ₹525.00. So both mean and median become ₹525 when W7 is removed

4.Total wage bill for all 7 workers=₹4,350.

5) If “Standard Wage” = median (₹550): how many earn below and above it?

Earn below ₹550: 400, 450, 500 → 3 workers.

Earn equal to ₹550: 550 → 1 worker.

Earn above ₹550: 600, 650, 1200 → 3 workers.

34. (**Liberal Valuation**)

To find the Present Value (PV) of a future amount, we use the Present Value formula:

$PV=F(1+r)^n$

where:FV=25,000 (Future Value), r=10%=0.10 (Discount rate),n=5 years

=15520.36

35. (**Liberal Valuation**)

Expression	Operator Type	Result
=30<35	Comparison	TRUE
=5*7	Arithmetic	35
=91/8	Arithmetic	11.375
=9-13	Arithmetic	-4
=50>=50	Comparison	TRUE
=40<=35	Comparison	FALSE
=AND(10>5,3<4)	Logical	TRUE
=OR(8>10,6<5)	Logical	FALSE
=90<>90	Comparison	FALSE
=100=100	Comparison	TRUE

Summary:

- Arithmetic Operators: *, /, -
- Comparison Operators: <, >, <=, >=, =, <>
- Logical Operators: AND(), OR()

36. (**Liberal Valuation)**.
The Internal Rate of Return (IRR) is a financial tool used to evaluate the profitability of investment projects. It represents the discount rate at which the Net Present Value (NPV) of a project's cash flows becomes zero. In simpler terms, IRR is the rate of return that makes the project break even in present value terms.

Evaluating Mutually Exclusive Projects:

When two or more projects are mutually exclusive (i.e., only one can be chosen), IRR helps compare their potential returns.

- A higher IRR generally indicates a more profitable project, provided other factors like project size and timing are similar.
- However, IRR should be used along with NPV, as IRR alone can sometimes give misleading results — especially when cash flows differ significantly in timing or scale.

The Multiple IRR Problem: The multiple IRR problem occurs when a project has non-conventional cash flows, meaning the signs of cash flows change more than once (for example, from positive to negative and back to positive).
Each change in sign can result in a different IRR value, making it unclear which rate is the correct one for decision-making.

(5x3=15 marks)

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